

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-7 Shared Preferences

Contents

1	Introduction	3
2	Activity Time boxing	3
3	Objectives of the experiment.....	3
4	Walk through Task	3
5	Home Task	7

1 Introduction

Android provides many ways of storing data of an application. One of this way is called Shared Preferences. Shared Preferences allow you to save and retrieve data in the form of key, value pair. The first parameter is the key and the second parameter is the MODE.

You can save something in the sharedPreferences by using SharedPreferences. Editor class. You will call the edit method of SharedPreferences instance and will receive it in an editor object. Its syntax is `Editor editor = sharedPreferences.edit(); editor.putString("key", "value"); editor.commit();`

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding Shared Preferences
- To get basic understanding regarding how data is saved and retrieved
- To get to know about the benefits related to Shared Preferences

4 Walk through Task

Creating New Activity

Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.

activity_my_prefs.xml

User Name

PassWord

Save

Get

Delete

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/user_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="User Name" />

    <EditText
        android:id="@+id/pass_word"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="PassWord" />

    <Button
        android:id="@+id/save_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="saveButton"
        android:text="Save" />

    <Button
        android:id="@+id/restore_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="restoreButton"
        android:text="Get" />

    <Button
        android:id="@+id/delete_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="deleteButton"
        android:text="Delete" />

</LinearLayout>

```

Step 3: Write shared preference class

```

import android.content.Context;
import android.content.SharedPreferences;

public class MySharedPrefs {

```

```

        public static final String USER_NAME = "userName";
        public static final String PASS_WORD = "passWord";

        public static boolean setUserNamePW(Context context, String userName, String
passWord) {
            SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
context.MODE_PRIVATE);
            SharedPreferences.Editor editor = prefs.edit();

            editor.putString(USER_NAME, userName);
            editor.putString(PASS_WORD, passWord);
            return editor.commit();
        }
        public static String[] getUserNamePW(Context context) {
            SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
context.MODE_PRIVATE);

            String[] userNamePW = new String[2];
            userNamePW[0] = prefs.getString(USER_NAME, "");
            userNamePW[1] = prefs.getString(PASS_WORD, "");
            return userNamePW;
        }
        public static boolean removeUserNamePW(Context context) {
            SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
context.MODE_PRIVATE);
            SharedPreferences.Editor editor = prefs.edit();

            editor.remove(USER_NAME);
            editor.remove(PASS_WORD);
            return editor.commit();
        }
    }
}

```

Step 4: Write main activity class

```

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import com.hitecuni.dce.lab4intent.R;

public class PrefActivity extends AppCompatActivity {

    EditText userName, password;
    Button save, restore, delete;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_my_prefs);

        userName = findViewById(R.id.user_name);
        password = findViewById(R.id.pass_word);
        save = findViewById(R.id.save_btn);
        restore = findViewById(R.id.save_btn);
        delete = findViewById(R.id.delete_btn);
    }
}

```

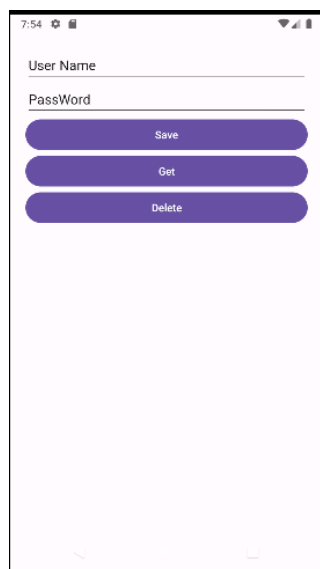
```

    }
    public void saveButton(View view){
        String userNameStr = userName.getText().toString();
        String passWordStr = password.getText().toString();
        if(!userNameStr.isEmpty() &&
            !passWordStr.isEmpty()){
            if(MySharedPrefs.setUserNamePW(this, userNameStr, passWordStr)){
                Toast.makeText(this, "Successfully Saved",
                    Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Not Successfully Saved",
                    Toast.LENGTH_SHORT).show();
            }
        }
    }
    public void restoreButton(View view){
        String[] userNamePW = MySharedPrefs.getUserNamePW(this);
        if(!userNamePW[0].isEmpty() && !userNamePW[1].isEmpty()){
            userName.setText(userNamePW[0]);
            password.setText(userNamePW[1]);
        }
    }
    public void deleteButton(View view){
        if(MySharedPrefs.removeUserNamePW(this)){
            userName.setText("");
            password.setText("");
            Toast.makeText(this, "Successfully Deleted",
                Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Not Successfully Deleted",
                Toast.LENGTH_SHORT).show();
        }
    }
}

```

Step 6: Now Run Project

Click “Run” or press “Shift + F10”.



5 Home Task

Use Tasbeeh app and use shared preference to store history.

Note: Shared Preference can only store strings.